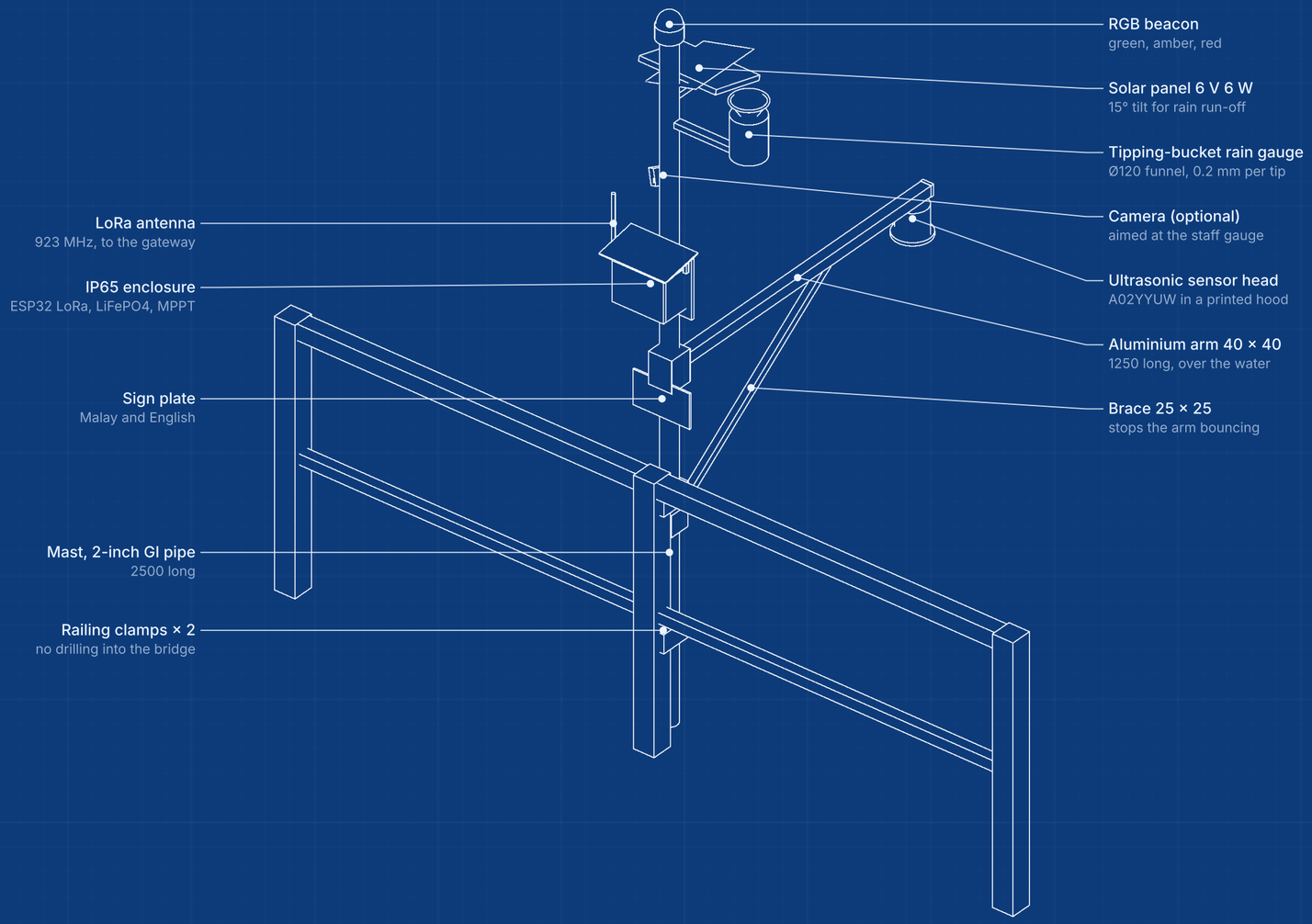
General arrangement

FloodWay RiverNode v1 clamped to a bridge railing, arm over the river



1 ISOMETRIC VIEW
from the bridge deck; railing shown for context

What it is

A solar river-level sensor that bolts onto a township bridge

It measures the river every few minutes, warns drivers on the bridge with its own beacon, and feeds FloodWay's alerts and safe routes.

MEASURES	river level, rate of rise, rainfall, air temperature
LEVEL SENSOR	A02YYUW ultrasonic, 3–450 cm, IP67, 60° beam
MOUNT	clamps to a bridge railing; arm 1.25 m over the water
CONTROLLER	ESP32 LoRa board (Heltec WiFi LoRa 32 V3)
RADIO	LoRa 923 MHz (AS923) to a gateway at the guardhouse
POWER	6 V 6 W solar panel, LiFePO4 3.2 V 6 Ah cell
LOCAL WARNING	RGB beacon on top, visible to drivers
CAMERA	optional ESP32-CAM, reads the staff gauge
PARTS COST	RM 560–950 per node (estimate, see S4)

Sheets: S2 elevations · S3 plan and install · S4 exploded view and parts
S5 details · S6 wiring · S7 firmware and power · S8 how it joins FloodWay

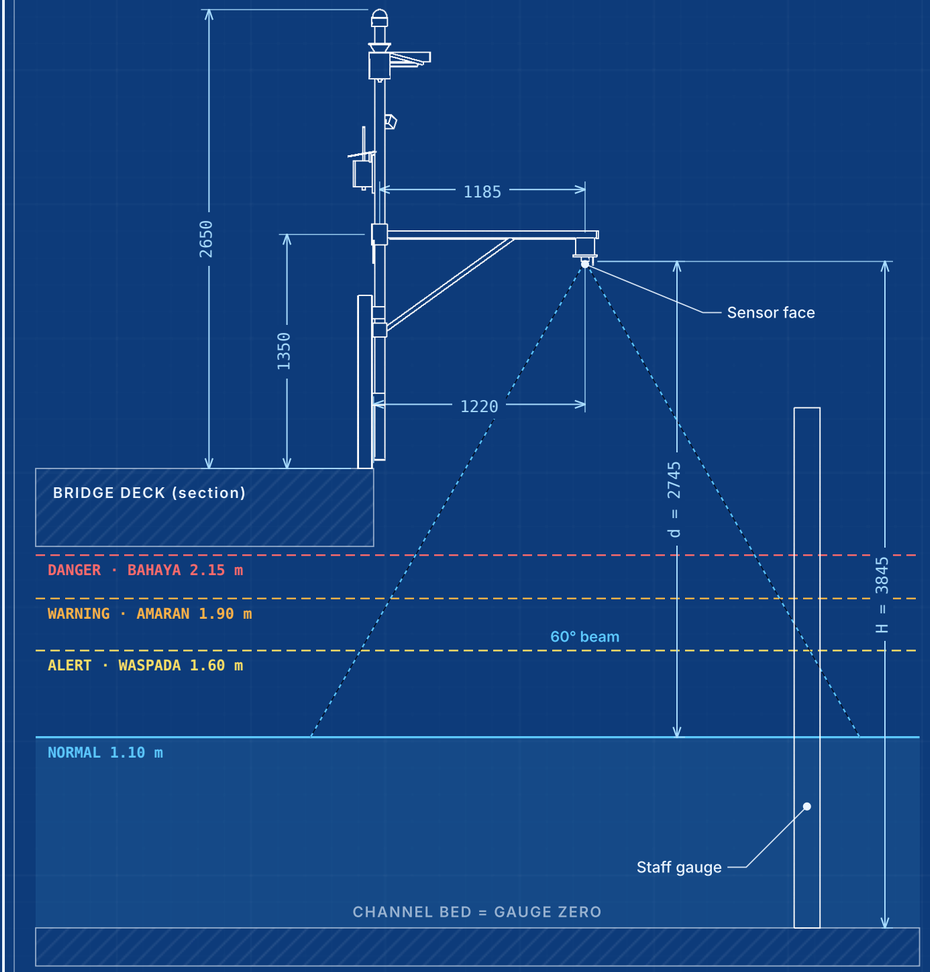
Concept design for pitching and a student build. Not for construction. Check every dimension on site before cutting or drilling.

FLOODWAY RIVERNODE v1 · FLOOD LEVEL SENSOR

SHEET TITLE GENERAL ARRANGEMENT	SHEET S1 of 8	
EVENT SDP University Hackathon 2026 · Township of Tomorrow Drawn: FloodWay team, UiTM Shah Alam	UNITS mm	REV · DATE A concept 20.10.2026

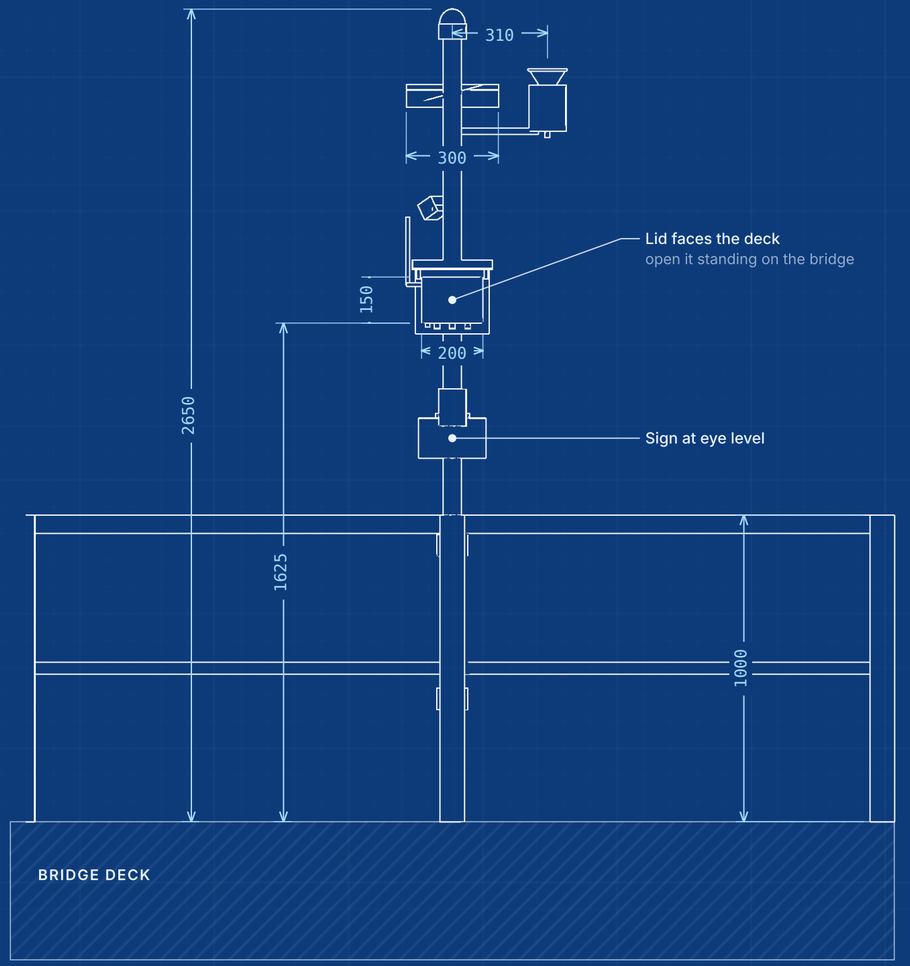
Elevations

Side and front elevations with mounting heights, measuring geometry and alert levels



A **SIDE ELEVATION**
looking across the river; arm points downstream

0 1 m



B **FRONT ELEVATION**
seen from the bridge deck

0 0.5 m

HOW IT MEASURES

The sensor times an ultrasonic echo off the water surface and turns it into a level:

$$d = c \times t \div 2$$

$$c = 331.3 + 0.606 \times T \text{ (m/s, T in °C)}$$

$$h = H - d$$

H is surveyed at install: sensor face to gauge zero. Here H = 3845 mm, so d = 2745 mm reads h = 1.10 m (normal).

EXAMPLE LEVELS (gauge reading)

Bahaya / Danger	2.15 m
Amaran / Warning	1.90 m
Waspada / Alert	1.60 m
Normal	1.10 m

Set the real marks with the council or JPS for the chosen river; these are examples.

BEAM CHECK

Nothing but water may sit inside the 60° cone. The cone edge passes the deck soffit 270 mm clear (36.6° > 30°).

READABLE RANGE

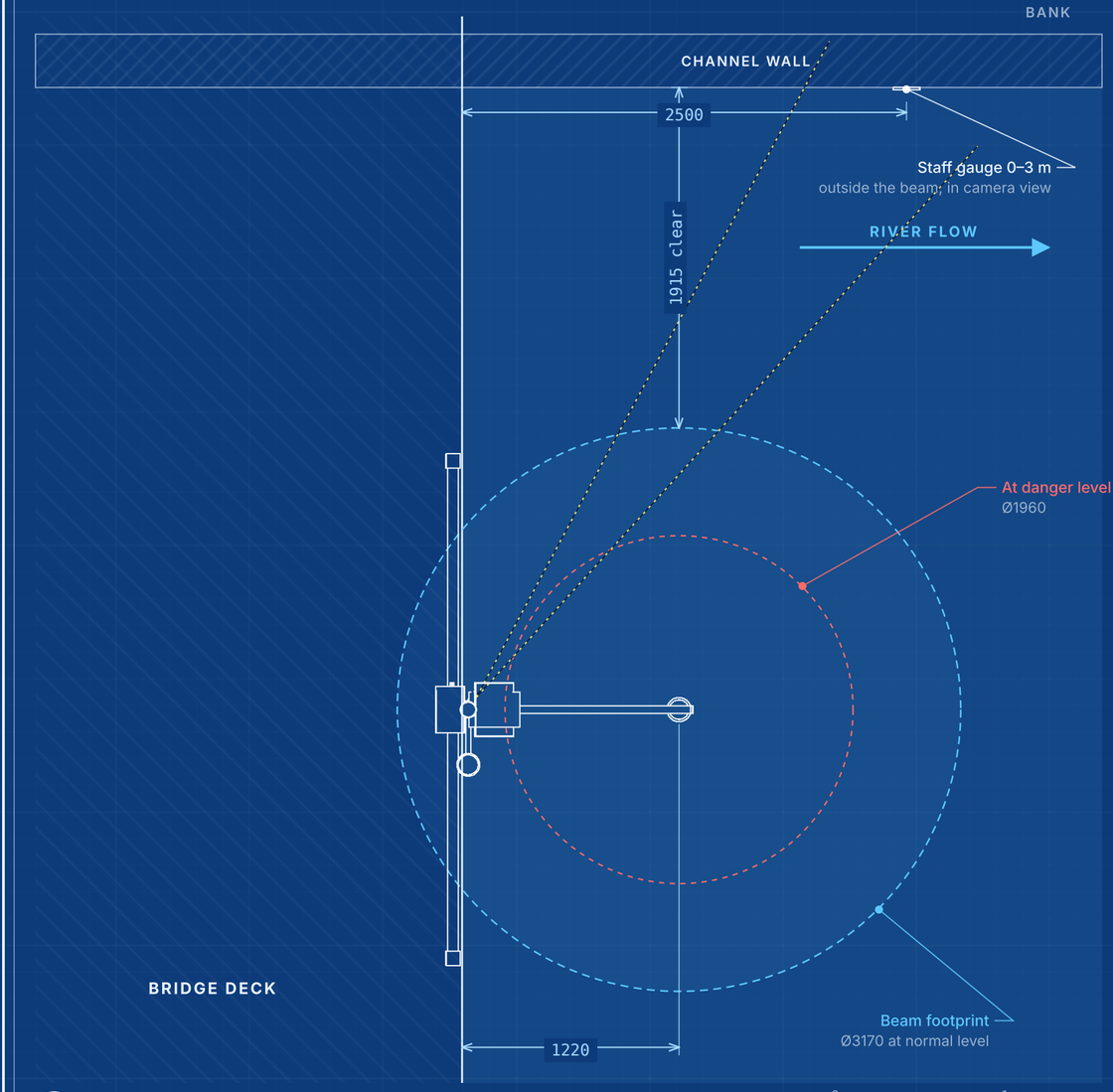
A02YYUW: 3–450 cm. Here d runs from 3845 (empty channel) to 1245 (water at deck level), all inside the range.

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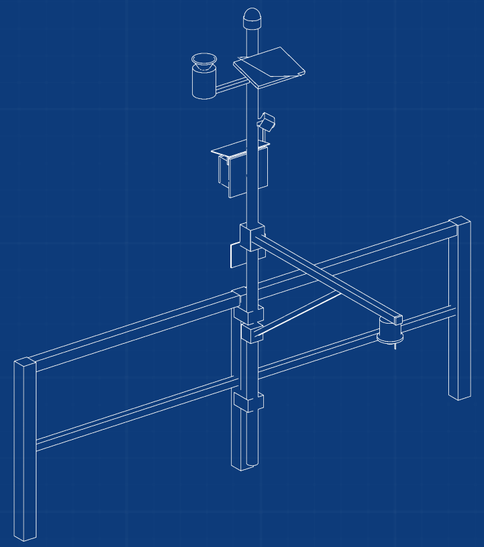
FLOODWAY RIVERNODE v1 · FLOOD LEVEL SENSOR		
SHEET TITLE ELEVATIONS	SHEET S2 of 8	
EVENT SDP University Hackathon 2026 · Township of Tomorrow Drawn: FloodWay team, UiTM Shah Alam	UNITS mm	REV · DATE A concept 20.10.2026

Plan and installation

Where the node sits on the bridge, what the beam sees, and the site rules

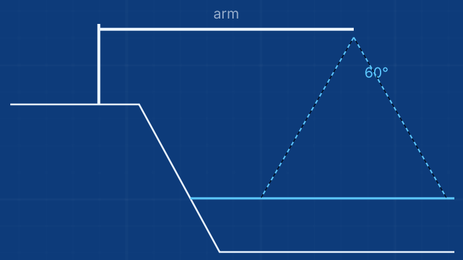


C PLAN
looking down; deck on the left, river flows left to right



D ISOMETRIC
from the river side

VARIANT B · RIVERBANK POST



Arm must reach d × tan 30° past the waterline:
about 1.7 m at d = 2.9 m. Use the bridge mount if possible.

INSTALLATION RULES

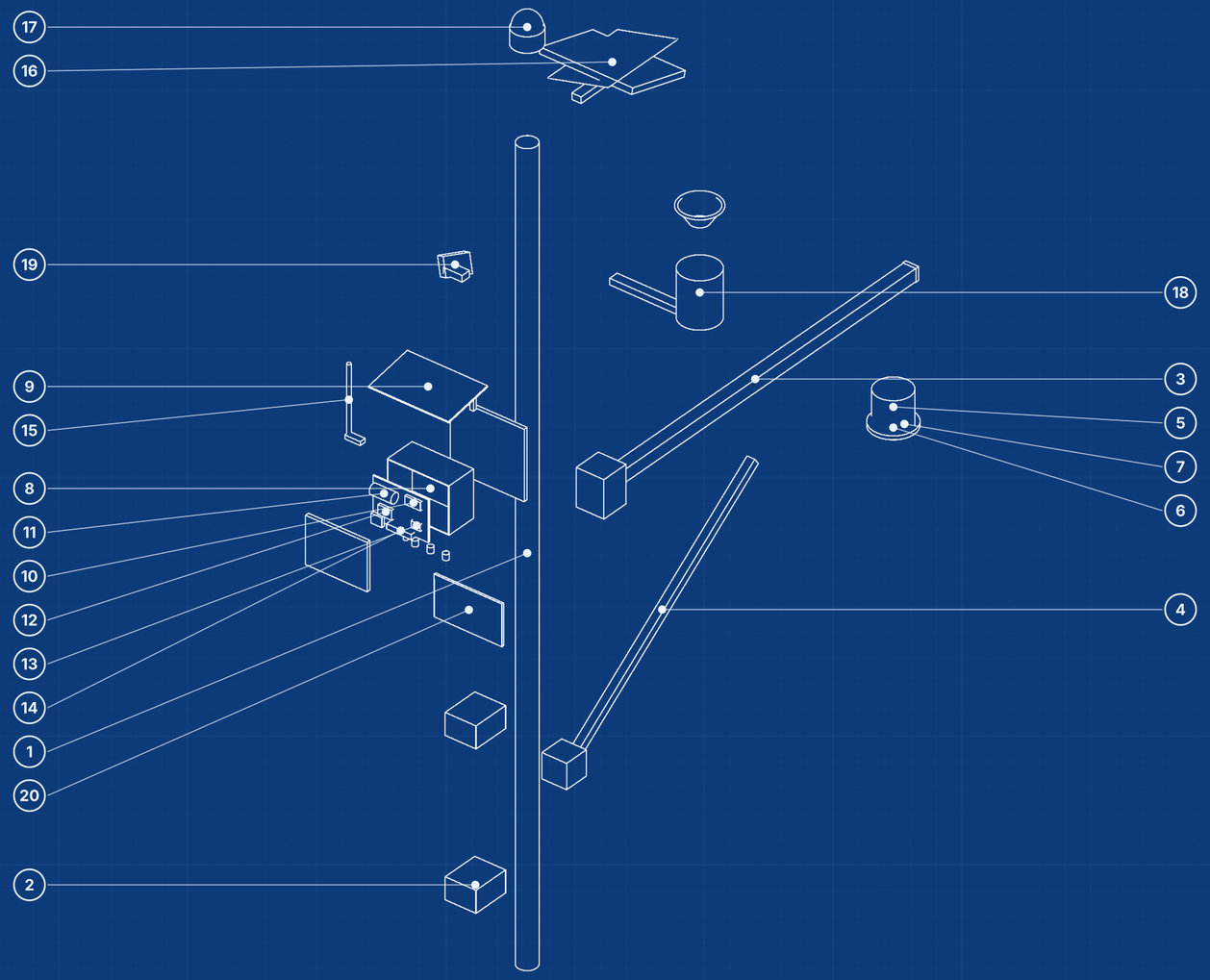
1. Pick a bridge on the township's flood route; fix to the downstream railing, mid-span, so debris hits the bridge first.
2. Keep the 60° cone clear: only water inside it at the lowest level. Here the arm reaches 1220 past the deck edge.
3. Fix the staff gauge to the channel wall, outside the beam and in view of the camera or township CCTV.
4. Survey the gauge zero and the sensor face at install; enter H (sensor face to gauge zero) in the firmware.
5. Get written permission from the bridge owner (local council or JKR) before clamping anything.
6. Earth the mast. Keep the box above the highest recorded flood level at the site.
7. Test LoRa range to the gateway before final fixing.

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FLOODWAY RIVERNODE v1 · FLOOD LEVEL SENSOR		
SHEET TITLE PLAN AND INSTALLATION	SHEET S3 of 8	
EVENT SDP University Hackathon 2026 · Township of Tomorrow Drawn: FloodWay team, UiTM Shah Alam	UNITS mm	REV · DATE A concept 20.10.2026

Exploded view and parts list

Every part a student team buys, prints or cuts, with an estimated cost



NO	PART	SPECIFICATION	QTY	EST. RM
1	**Mast**	2-inch GI pipe, 60.3 mm OD, 2500 long	1	60–90
2	**Railing clamps**	aluminium or stainless U-bolt clamp	2	30–50
3	**Arm**	aluminium square tube 40×40×2, 1250; clamp; end cap	1	40–70
4	**Brace**	aluminium square tube 25×25×2, 900; clamp	1	20–35
5	**Sensor hood**	3D-printed rPETG hood and drip skirt	1	10–20
6	**Ultrasonic sensor**	DFRobot A02YYUW, IP67, 3–450 cm, UART	1	80–120
7	**Air temperature probe**	DS18B20 waterproof probe	1	8–15
8	**Enclosure**	IP65 ABS 200×150×100, backplate, 3 glands, vent	1	35–60
9	**Enclosure hood**	3D-printed or aluminium sheet rain hood	1	10–20
10	**Controller**	Heltec WiFi LoRa 32 V3 (ESP32-S3, AS923)	1	70–110
11	**Battery**	LiFePO4 32700 3.2 V 6 Ah, holder, 1S protection	1	35–55
12	**Solar charger**	CN3791 MPPT module set for LiFePO4	1	10–20
13	**Power modules**	3.3 V buck-boost and 5 V boost	1	10–20
14	**Terminals and fuse**	terminal blocks, 1 A fuse, silica gel	1	10–20
15	**LoRa antenna**	923 MHz 3 dBi, pigtail, bracket	1	15–30
16	**Solar panel**	6 V 6 W panel and bracket	1	30–60
17	**Beacon**	WS2812 12-LED ring, printed base, diffuser dome	1	15–30
18	**Rain gauge**	printed tipping bucket, Ø120 funnel, reed switch	1	15–25
19	Camera (optional)	ESP32-CAM, housing, bracket	1	30–50
20	**Sign plate**	printed vinyl on aluminium, bilingual	1	15–25
21	**Staff gauge**	0–3 m printed board and wall fixings (not shown)	1	40–70
22	Gateway	second ESP32 LoRa board at the guardhouse (not shown)	1	80–120

Node total (items 1–18, 20, 21) RM 558–945

Optional camera (19) and the shared gateway (22) are extra. Prices are estimates for Malaysia, October 2026, not quotes: check Cytron, Shopee or Lazada. Excludes tools, printer time and labour.

E **EXPLODED ISOMETRIC**
parts pulled apart along their fixing direction; numbers match the parts list

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FLOODWAY RIVERNODE v1 · FLOOD LEVEL SENSOR

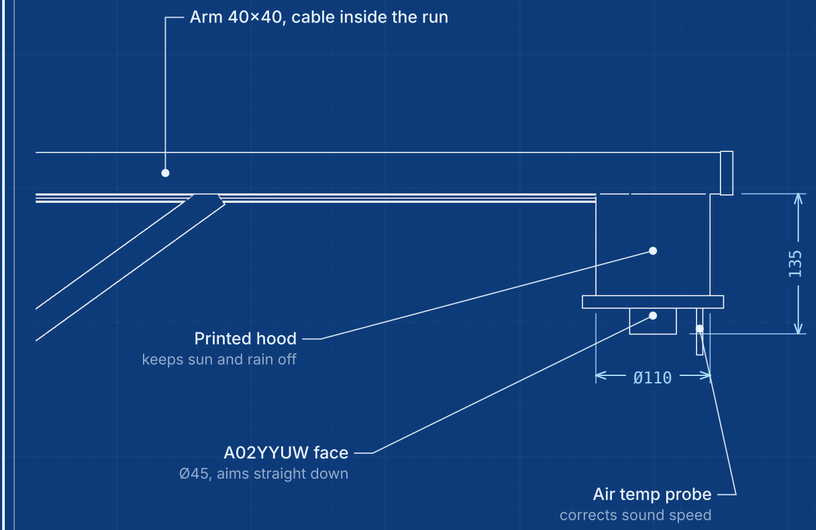
SHEET TITLE	SHEET	
EXPLODED VIEW AND PARTS LIST	S4 of 8	
EVENT SDP University Hackathon 2026 · Township of Tomorrow Drawn: FloodWay team, UiTM Shah Alam	UNITS mm	REV · DATE A concept 20.10.2026

Details

Sensor head, enclosure interior and rain gauge

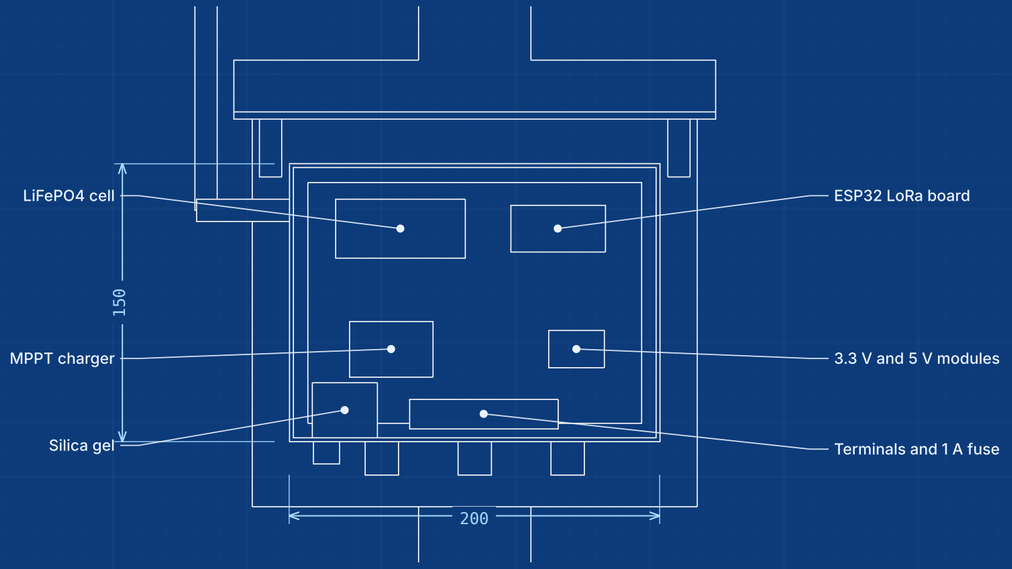
F SENSOR HEAD

side view, about 1:5



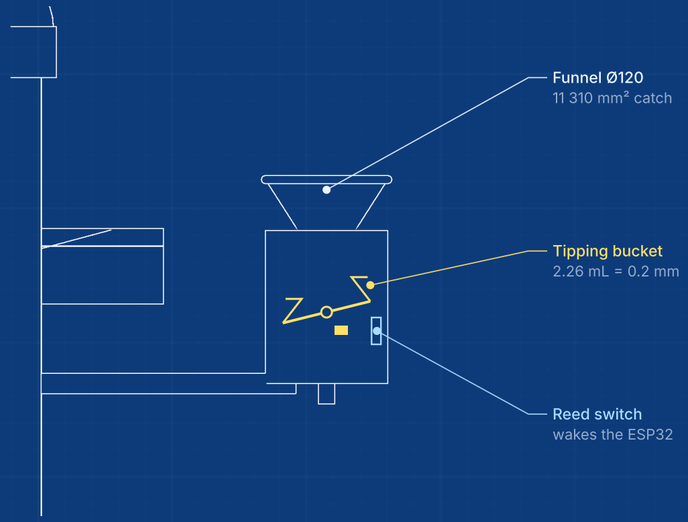
G ENCLOSURE, LID OFF

seen from the deck, about 1:3



H RAIN GAUGE

front view with the inside drawn, about 1:4



SENSOR HEAD NOTES

The A02YYUW blind zone is 3 cm, so the water may rise to 30 mm below the face and still read.
The probe sits beside the beam, not in it.
Seal the cable gland under the arm; drill a 3 mm drain hole at the low end of the hood.

ENCLOSURE NOTES

Heaviest part (battery) mounts low on the plate.
LiFePO4 is chosen because it tolerates heat in a sun-baked box better than Li-ion cells.
The breather vent stops the box pumping in moist air as it heats and cools; replace the silica gel yearly.

RAIN GAUGE NOTES

0.2 mm a tip: bucket volume = 11 310 mm² × 0.2 mm = 2262 mm³ = 2.26 mL.

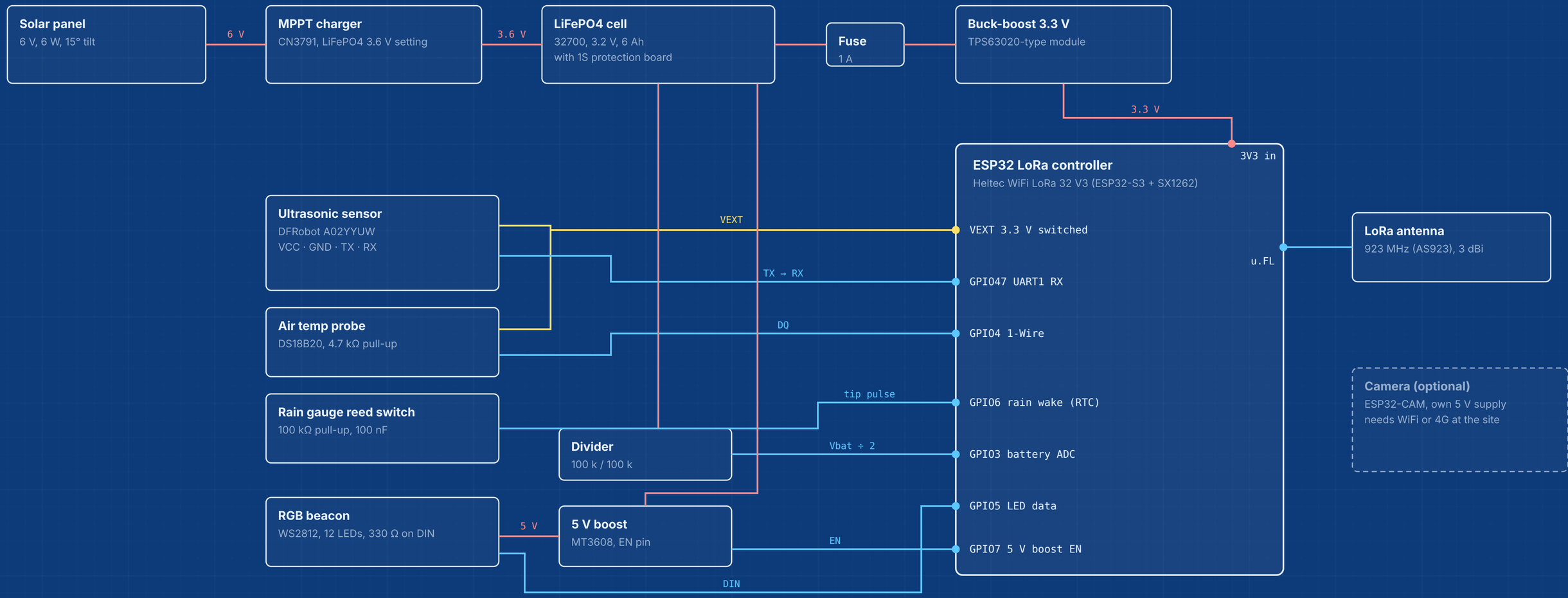
Calibrate: pour 113 mL (10 mm of rain) slowly through the funnel; count 50 tips. Adjust the end-stop screws until it does.

FLOODWAY RIVERNODE v1 · FLOOD LEVEL SENSOR		
SHEET TITLE DETAILS	SHEET S5 of 8	
EVENT SDP University Hackathon 2026 · Township of Tomorrow Drawn: FloodWay team, UiTM Shah Alam	UNITS mm	REV · DATE A concept 20.10.2026

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Electronics and wiring

How the parts connect inside and outside the IP65 box



LEGEND

- Battery and supply
- Switched 3.3 V, off while asleep
- Signal

All GND pins join at the terminal block (not drawn). Pin numbers are suggestions for the Heltec V3: check its pinout before wiring.
Leave the A02YYUW RX pin as its datasheet says for the output mode you want. Feed the board at 3V3, not through its own battery socket.

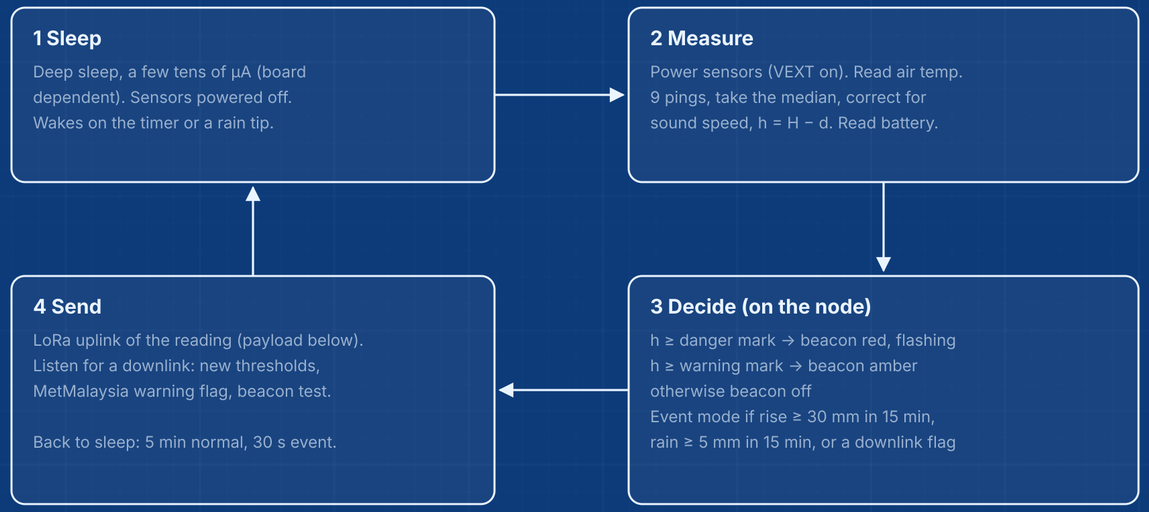
FLOODWAY RIVERNODE v1 · FLOOD LEVEL SENSOR		
SHEET TITLE ELECTRONICS AND WIRING	SHEET S6 of 8	
EVENT SDP University Hackathon 2026 · Township of Tomorrow Drawn: FloodWay team, UiTM Shah Alam	UNITS mm	REV · DATE A concept 20.10.2026

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Firmware and power

What the node does every wake-up, and how long it runs on its battery

FIRMWARE LOOP



PAYLOAD (one reading)

```
{"id":"BBR-01","ts":1792502400,
"level_m":1.10,"rise15_mm":4,
"rain15_mm":0.0,"air_c":29.5,
"batt_v":3.28,"mode":"NORMAL","beacon":"OFF"}
```

The node decides the beacon itself, so it still warns drivers if the radio or the internet is down.
The backend combines the reading with the other three inputs (S8) for the app and the dashboard.

POWER BUDGET (design estimate, measure on the bench)

MODE	WAKE EVERY	AVG CURRENT	ENERGY / DAY	DAYS WITHOUT SUN
Normal	5 min	≈1 mA	≈0.1 Wh	≈190 (theory)
Event	30 s	≈10 mA	≈0.9 Wh	≈20
Danger + beacon	30 s	+ ≈36 mA at 5 V	≈5.2 Wh	≈4

Assumes 3 s awake at about 100 mA per wake-up, 3.3 V rail at 85% efficiency, and a 19.2 Wh cell (3.2 V × 6 Ah). Beacon at 30% brightness, 50% duty.
Solar: 6 W × about 4 peak sun hours × 0.6 losses ≈ 14 Wh on a clear day; monsoon overcast days give much less, so the battery carries the gap.

WHY THESE CHOICES

Solar + LiFePO4: no mains cable on the bridge; LiFePO4 handles heat better.
LoRa, not WiFi: a few milliamps per message and no SIM fee; one gateway can serve several nodes within range (range to be measured on site).
Ultrasonic, not a submerged sensor: nothing touches the water, so debris and silt do not foul it, and the whole node can be serviced from the deck.
3D-printed parts in recycled PETG: cheap to replace and easy to repair.

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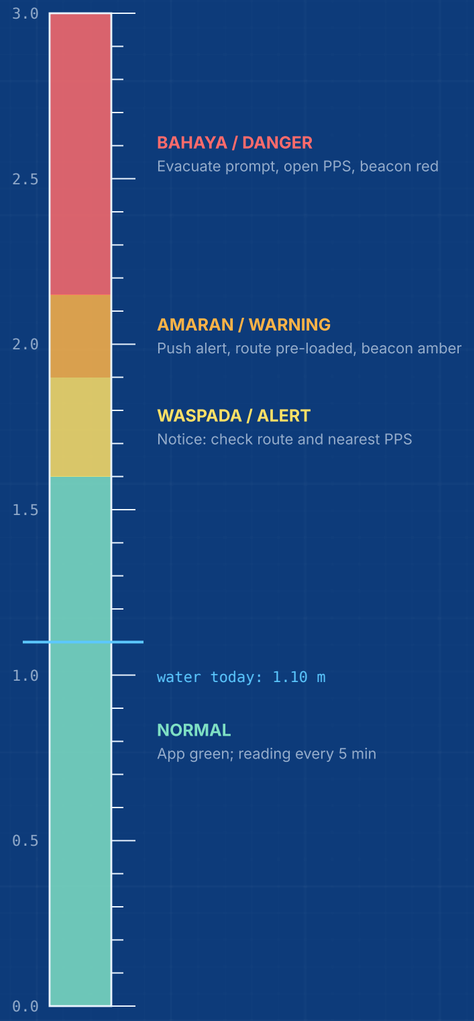
FLOODWAY RIVERNODE v1 · FLOOD LEVEL SENSOR

SHEET TITLE	SHEET	
FIRMWARE AND POWER	S7 of 8	
EVENT: SDP University Hackathon 2026 · Township of Tomorrow Drawn: FloodWay team, UiTM Shah Alam	UNITS: mm	REV · DATE: A concept 20.10.2026

How it joins FloodWay

The RiverNode is input 3 of the four that FloodWay combines into one alert level

STAFF GAUGE AND ALERT LEVELS (m)



HOW THE NODE JOINS THE FLOODWAY PREDICTION

RiverNode (this device)
level, rise, rain, air temp

LoRa 923 MHz

Gateway at the guardhouse
ESP32 LoRa + township WiFi

internet

FloodWay backend
FastAPI + Firebase; stores every reading

stored readings = input 3

1 **Trained models**
where floods are likely
GINKGO map + rainfall RF

2 **MetMalaysia API**
when heavy rain is coming
forecasts and warnings

3 **RiverNode**
what the river is doing now
this device, every 30 s–5 min

4 **CCTV**
what people would see
water over a marked line

Fusion engine (rules, not another model)

Any input raises the level one step. Danger needs two inputs to agree, or the RiverNode above its surveyed danger mark.

If the node is offline, the other three inputs carry on. Thresholds live in one config file, tuned during the pilot.

Resident app
alert and safe route to an open PPS

Beacon on the node
downlink: amber or red at the bridge

Township dashboard
SDP township team and the council

THE DATA LOOP

When a camera can see the staff gauge, every RiverNode reading labels that frame automatically, which trains and tests the CCTV check.

Each flood the node records becomes local ground truth for re-testing the trained models on Malaysian sites.

FLOODWAY RIVERNODE v1 · FLOOD LEVEL SENSOR		
SHEET TITLE HOW IT JOINS FLOODWAY	SHEET S8 of 8	
EVENT SDP University Hackathon 2026 · Township of Tomorrow Drawn: FloodWay team, UiTM Shah Alam	UNITS mm	REV · DATE A concept 20.10.2026

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